DNS: Ejemplo práctico

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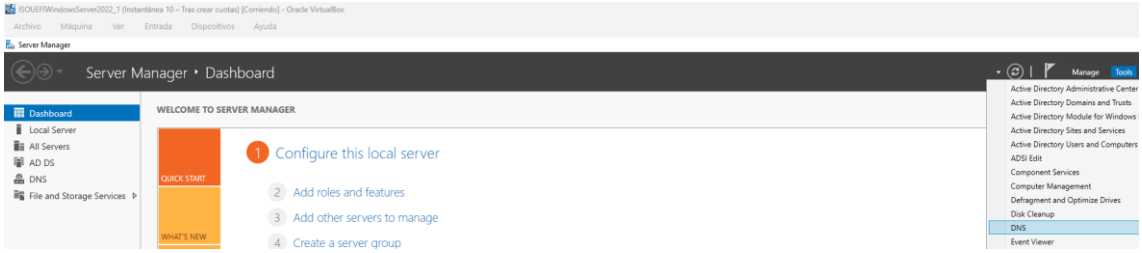
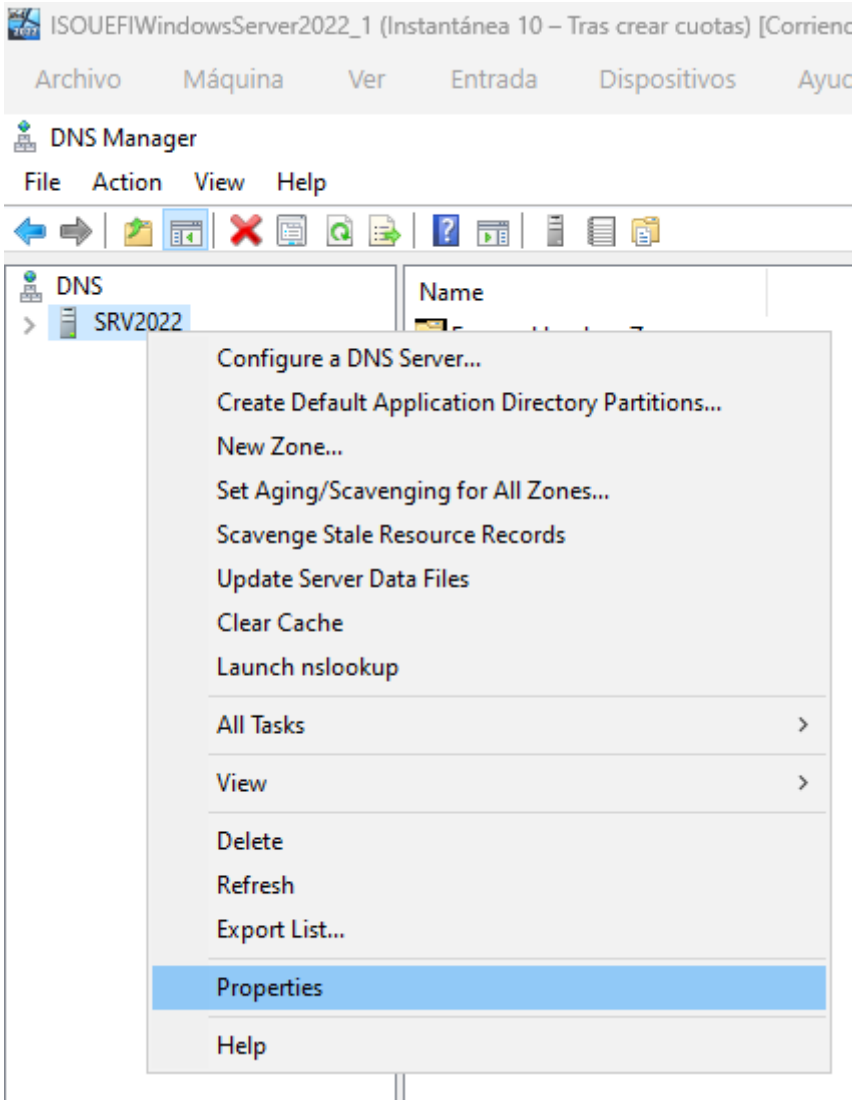
Configuración de la zona DNS

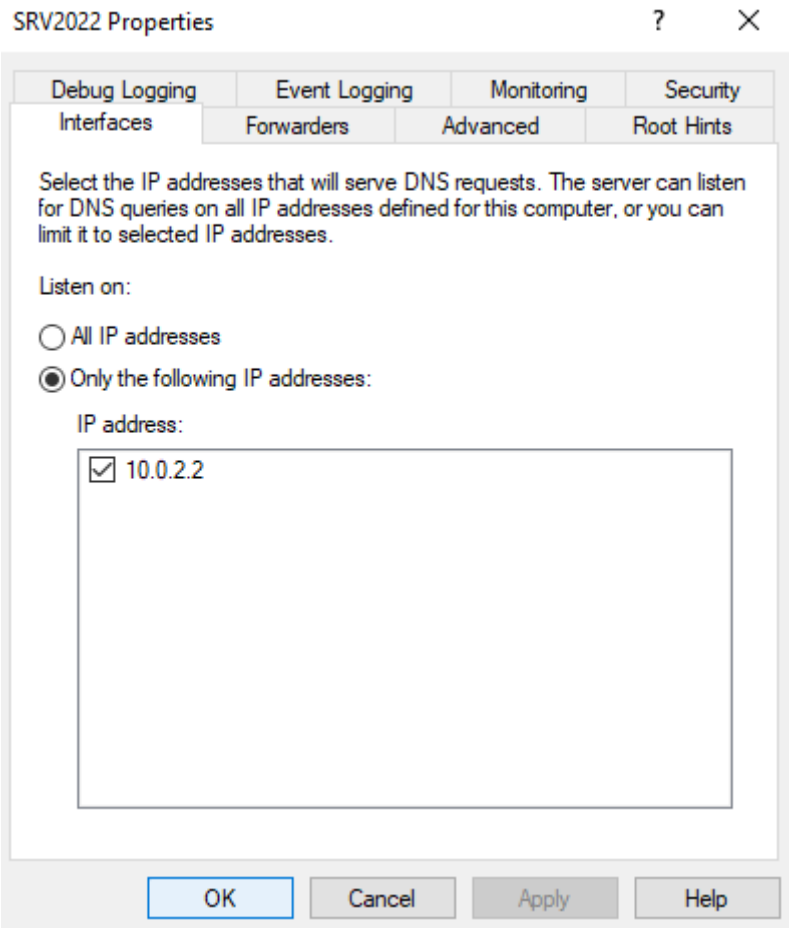
1. Encender la máquina virtual servidor “ISOUEFIWindowsServer2022_1”.
2. Loguear.
3. En “Server Manager”, pulsar en “Tools” y en el menú contextual pulsar en “DNS”:



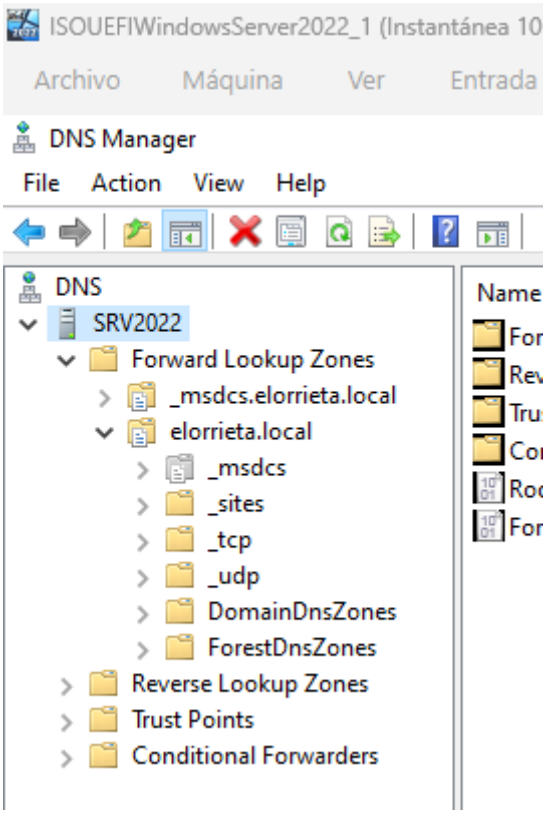
4. En “DNS Manager”, en el panel lateral izquierdo, pulsar “SRV2022”, click derecho en “SRV2022” y en el panel contextual pulsar “Properties”:



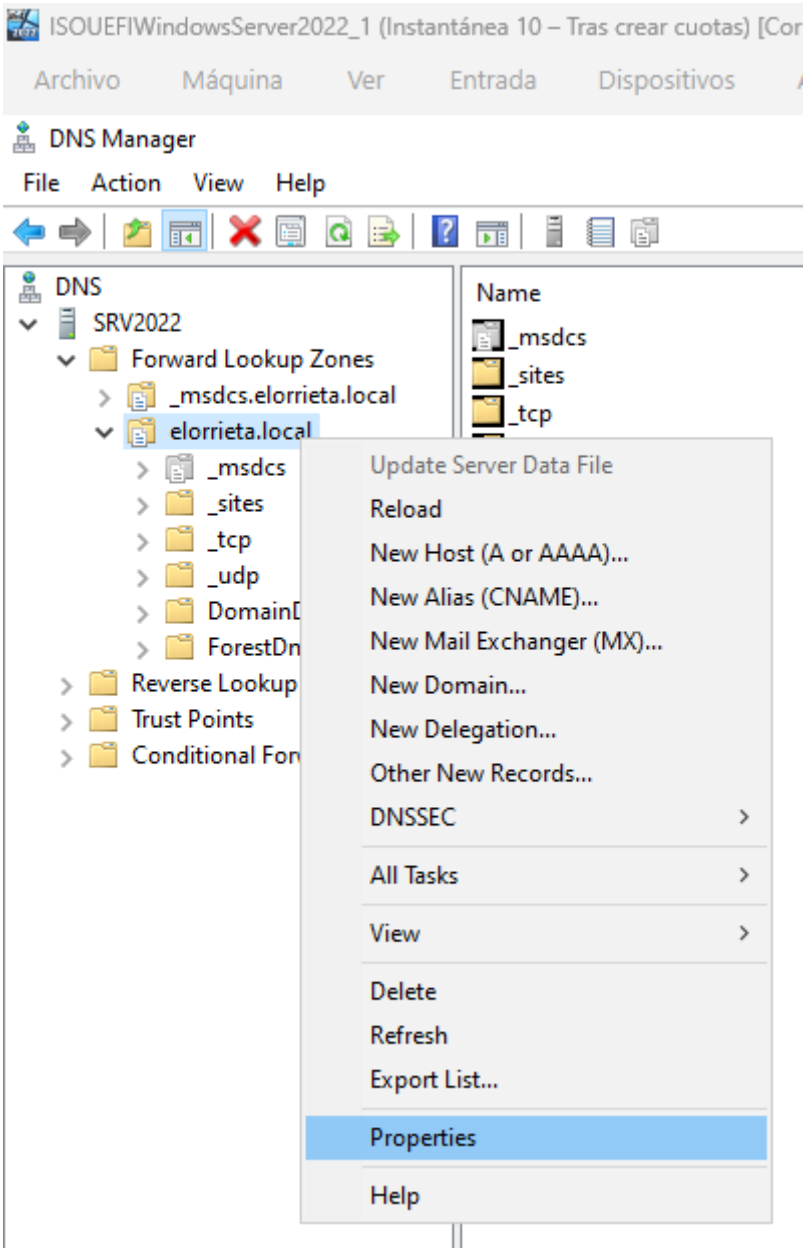
5. En “SRV2022 Properties”, en la pestaña “Interfaces”, pulsar “Only the following IP address:”, dejar en el listado de “IP address:” solo marcado el check de la IP 10.0.2.2, pulsar “Apply” y pulsar “OK”:



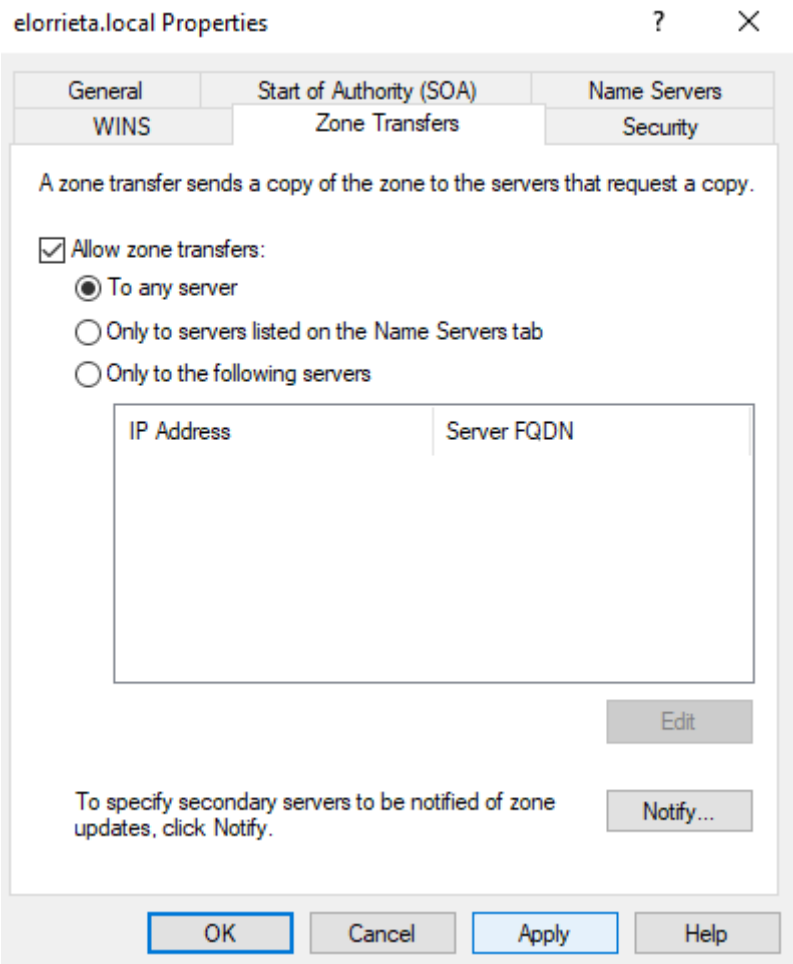
6. En “DNS Manager”, en el panel lateral izquierdo, desplegar “SRV2022”, desplegar “Forward Lookup Zones” y desplegar “elorrieta.local”:



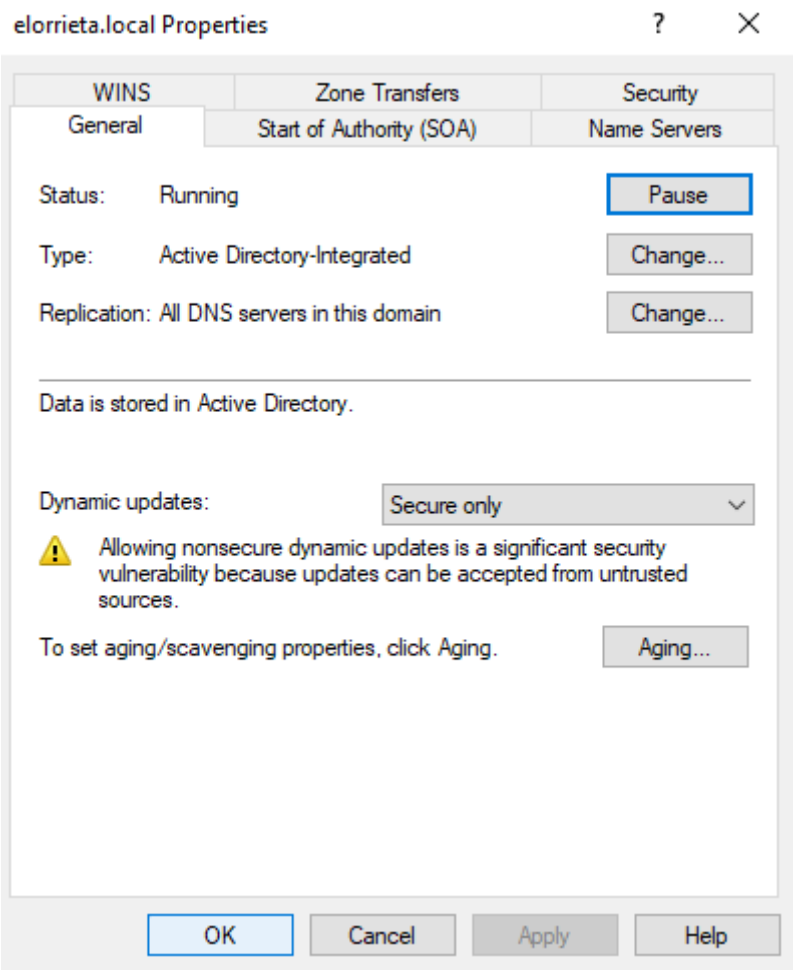
7. En “DNS Manager”, en el panel lateral izquierdo, pulsar “elorrieta.local”, click derecho en “elorrieta.local” y en el panel contextual pulsar “Properties”:



8. En “elorrieta.local Properties”, pulsar “Zone Transfers”, marcar el check “Allow zone transfers:” y pulsar “Apply”:

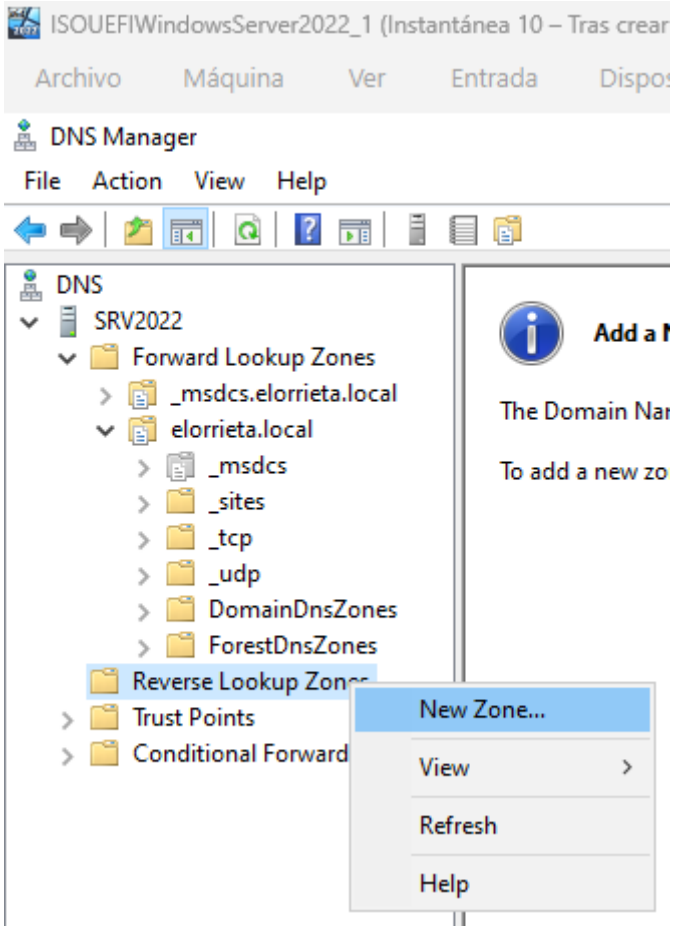


9. En “elorrieta.local Properties”, pulsar “General”, verificar que en “Type” aparece el valor “Active Directory-Integrated”, en “Replication:” aparece el valor “All DNS servers in this domain”, en “Dynamic updates:” aparece el valor “Secure only” y pulsar “OK”:


elorrieta.local Properties
WINS
Zone Transfers
Security
General
Start of Authority (SOA)
Name Servers
Status: Running
Pause
Type: Active Directory-Integrated
Change...
Replication: All DNS servers in this domain
Change...
Data is stored in Active Directory.
Dynamic updates:
Secure only
Allowing nonsecure dynamic updates is a significant security vulnerability because updates can be accepted from untrusted sources.
To set aging/scavenging properties, click Aging.
Aging...
OK
Cancel
Apply
Help

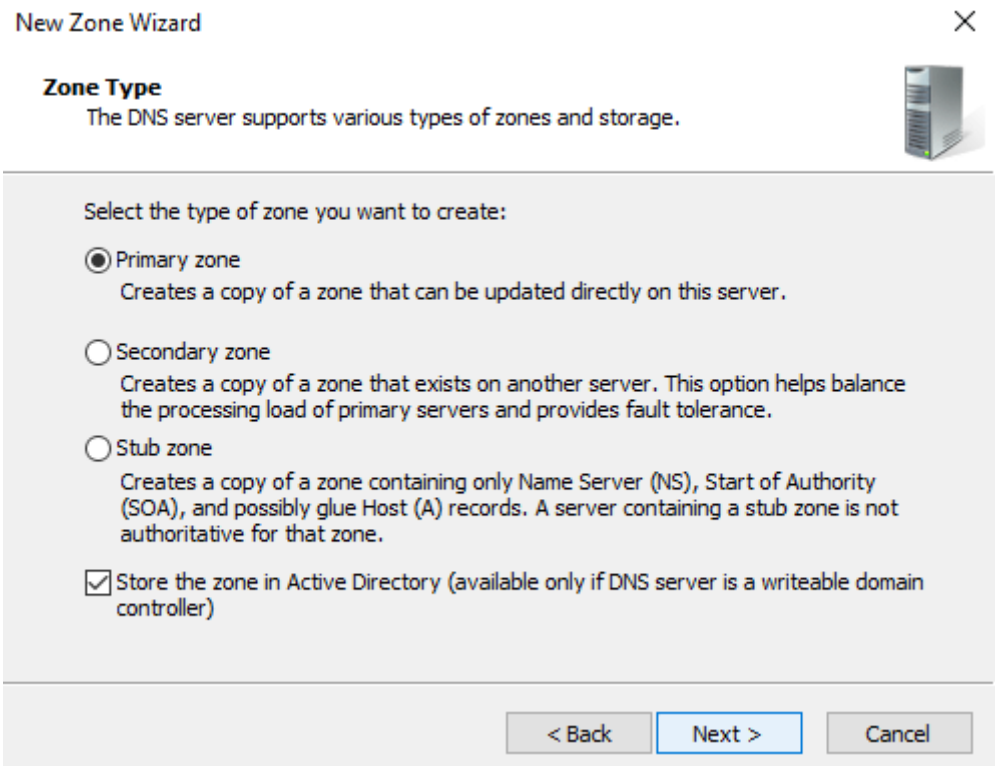
Creación de una zona de búsqueda inversa

1. En “DNS Manager”, en el panel lateral izquierdo, pulsar “Reverse Lookup Zones”, click derecho en “Reverse Lookup Zones” y en el panel contextual pulsar “New Zone ...”:

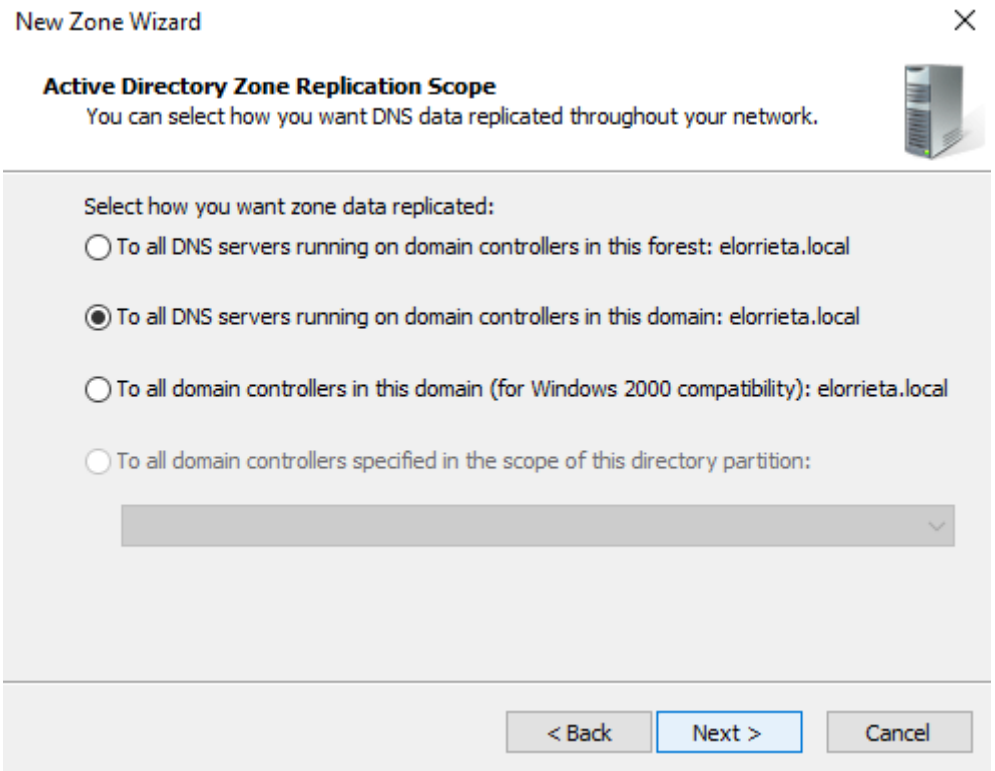


2. En “New Zone Wizard”, “Welcome to the New Zone Wizard”, pulsar “Next >”:

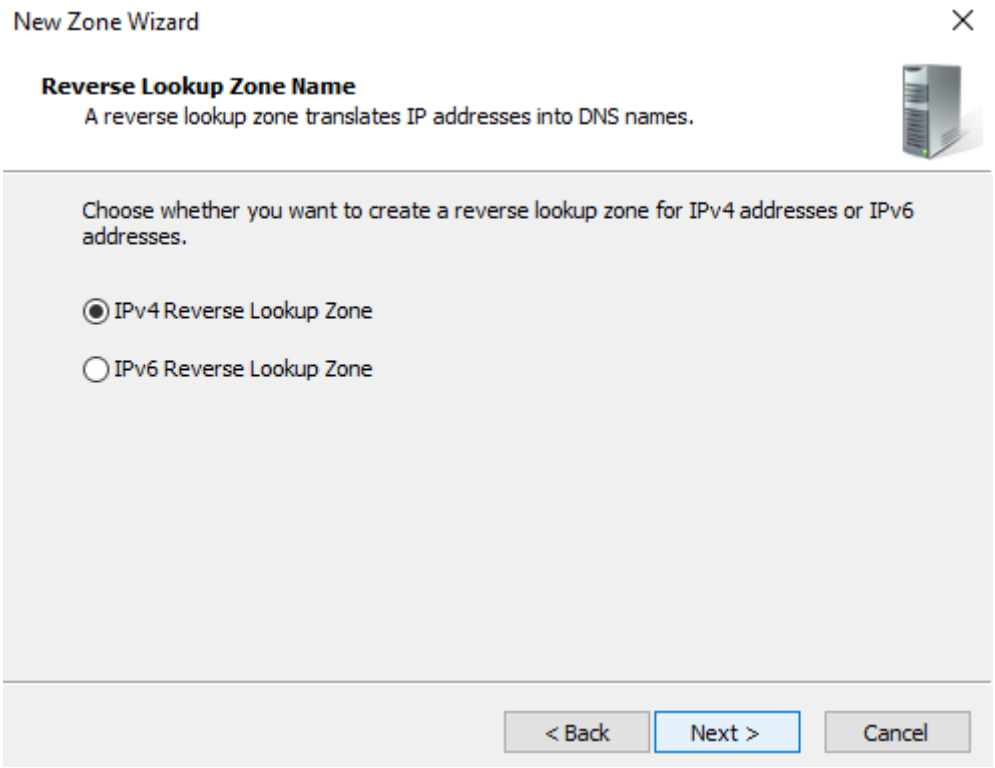
3. En “New Zone Wizard”, “Zone Type”, dejar marcada la opción “Primary zone”, dejar marcado el check “Store the zone in Active Directory (available only if DNS server is a writeable domain controller” y pulsar “Next >”:



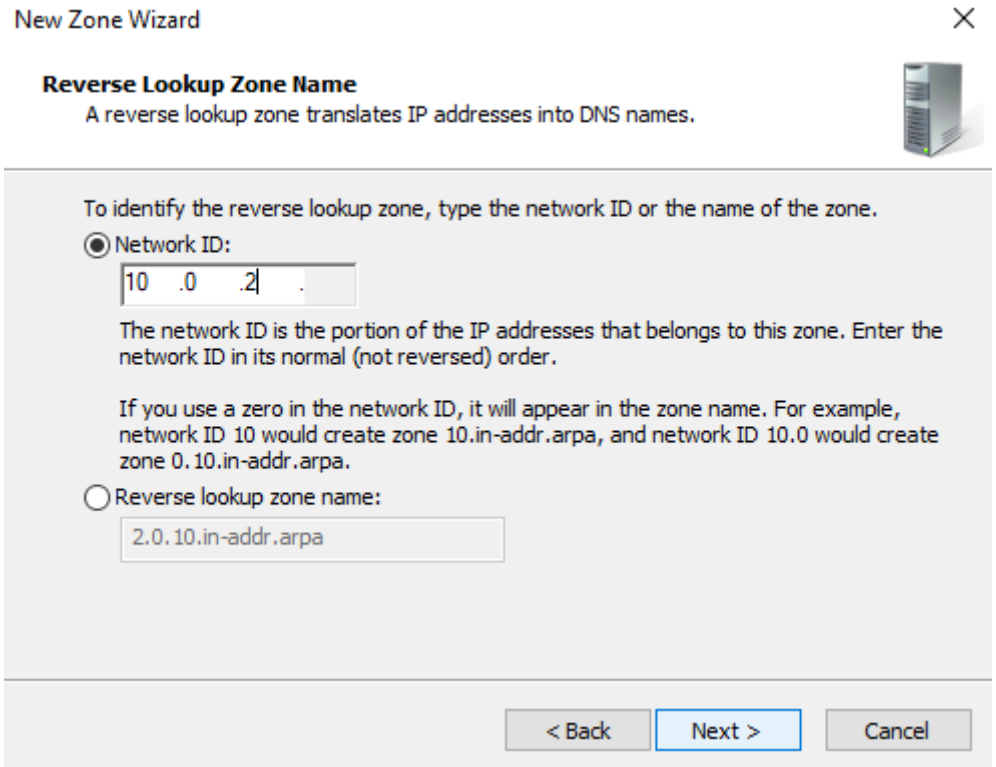
4. En “New Zone Wizard”, “Active Directory Zone Replication Scope”, dejar marcada la opción “To all DNS servers running on domain controllers in the domain: elorrieta.local” y pulsar “Next >”:



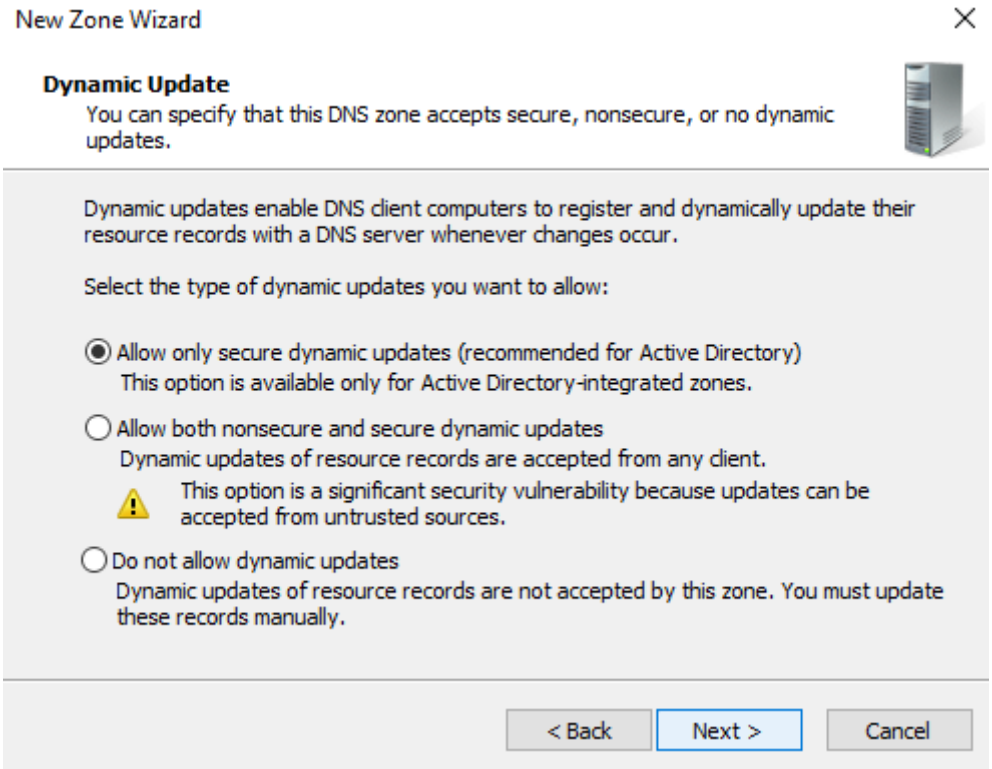
5. En “New Zone Wizard”, “Reverse Lookup Zone Name”, dejar marcada la opción “IPv4 Reverse Lookup Zone” y pulsar “Next >”:



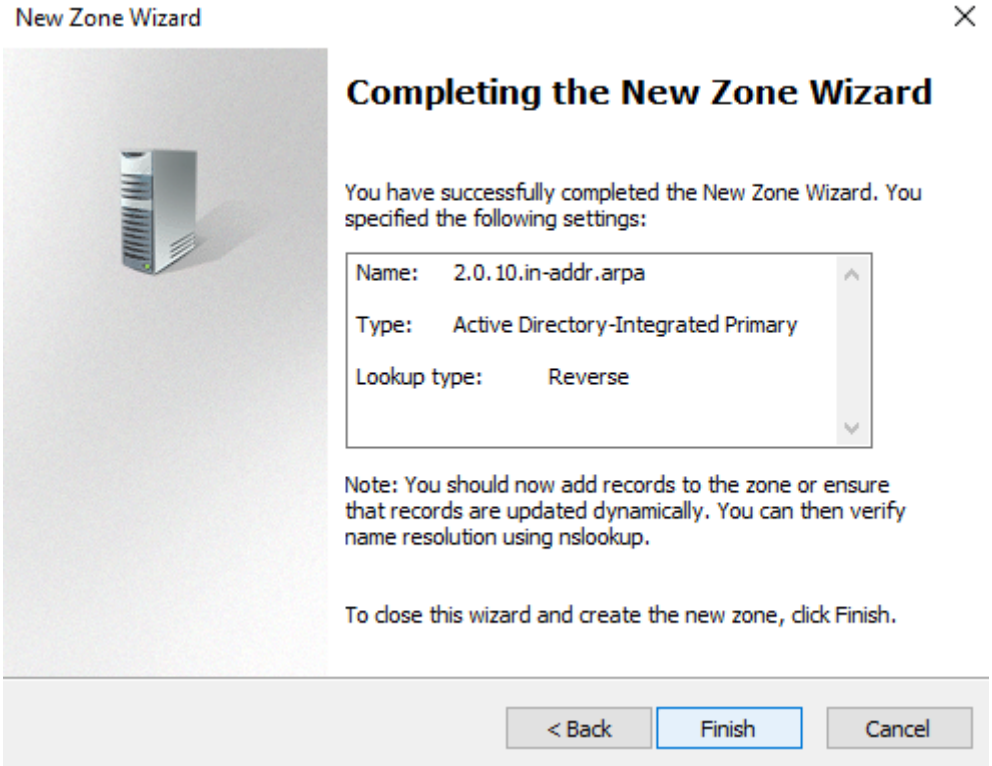
6. En “New Zone Wizard”, “Reverse Lookup Zone Name”, en “Network ID:”, escribir el ID de la red, que se puede deducir del resultado obtenible vía cmd seguido de “ipconfig /all”, observando la “IPv4 Address” y la “Subnet Mask”, y pulsar “Next >”:



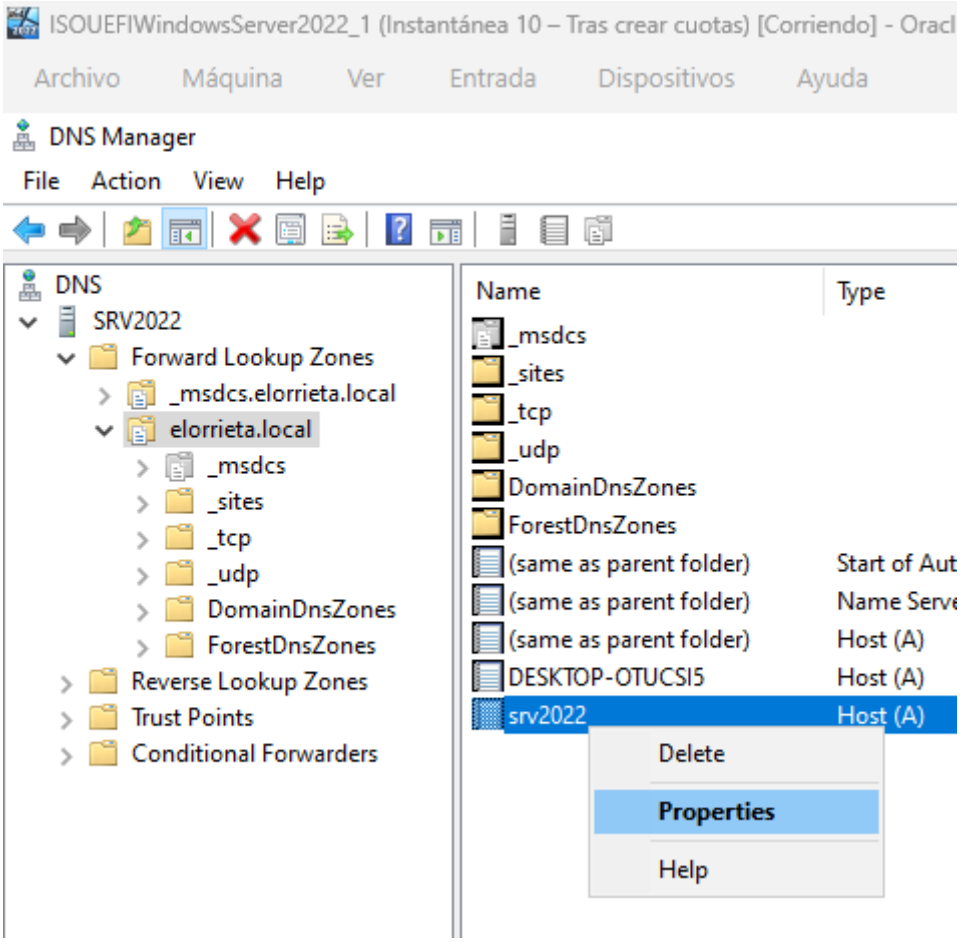
7. En “New Zone Wizard”, “Dynamic Update”, dejar marcada la opción “Allow only secure dynamic updates (recommended for Active Directory) This option is available only for Active Directory-integrated zones.” y pulsar “Next >”:



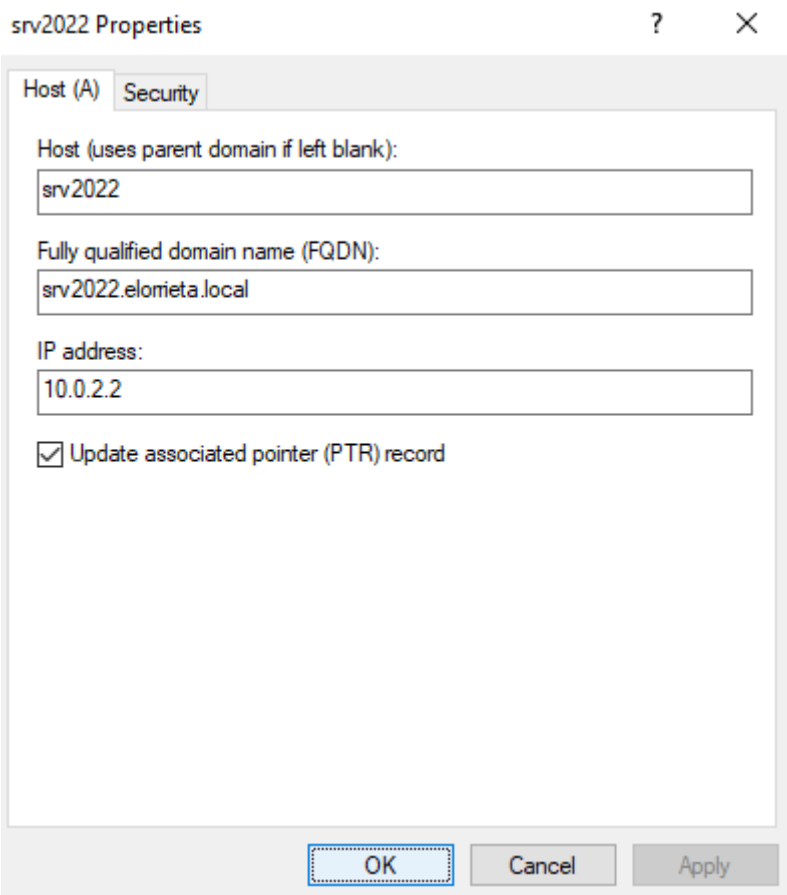
8. En “New Zone Wizard”, “Completing the New Zone Wizard”, pulsar “Finish”:



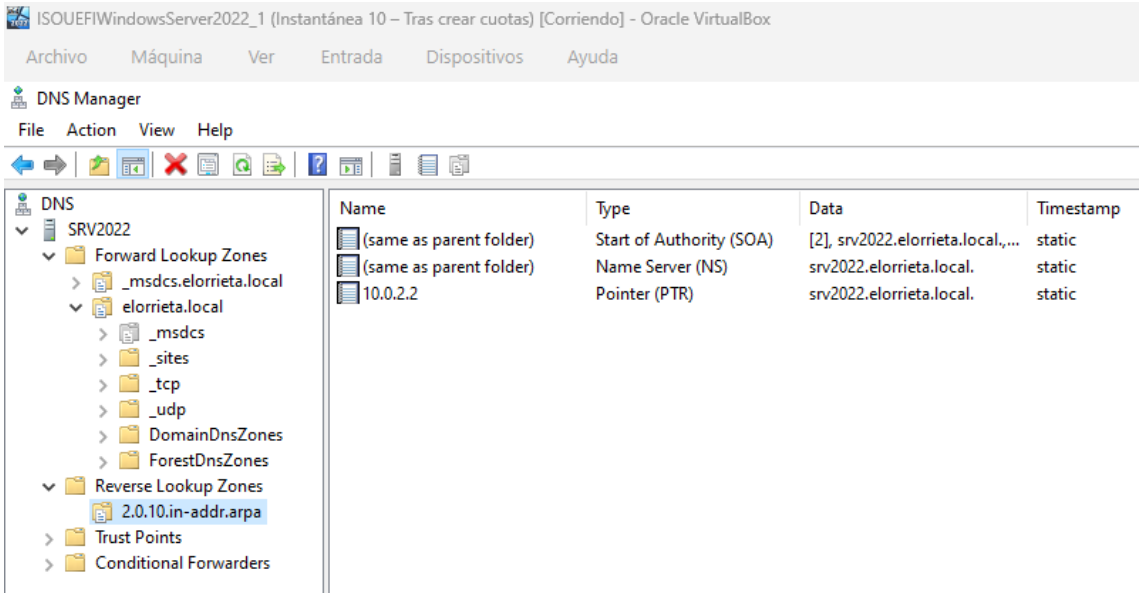
9. En “DNS Manager”, en el panel lateral izquierdo pulsar en “elorrieta.local”, en el panel central pulsar en “srv2022”, click derecho en “srv2022”, y en el panel contextual pulsar en “Properties”:



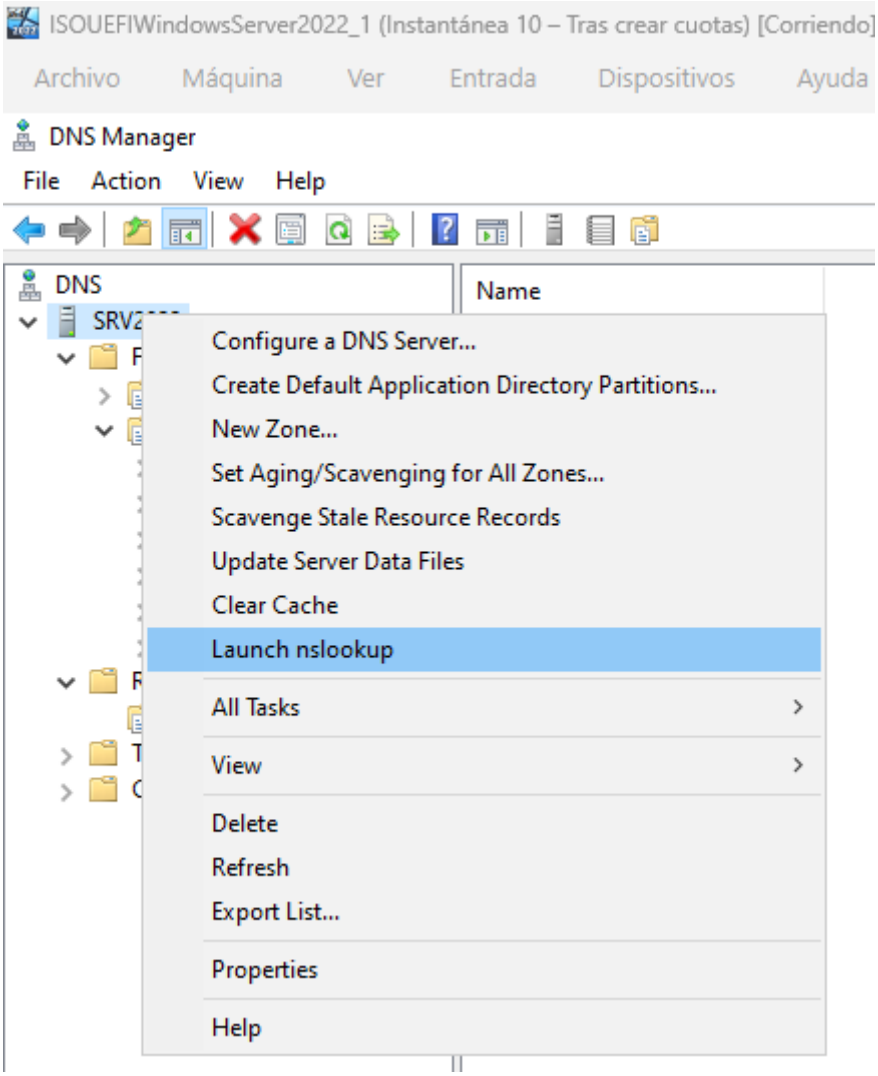
10. En “srv2022 Properties” marcar el check “Update associated pointer (PTR) record”, pulsar “Apply” y pulsar “OK”:



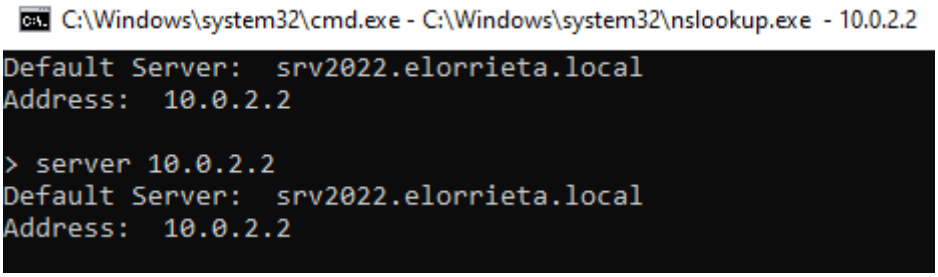
11. En “DNS Manager”, en el panel izquierdo desplegar “Reverse Lookup Zones” , pulsar en “2.0.10.in-addr.arpa” y verificar que en el panel central aparece el registro PTR:



12. En “DNS Manager”, en el panel lateral izquierdo pulsar en “SRV2022”, click derecho en “SRV2022” y en el panel contextual pulsar en “Launch nslookup”:



13. En “nslookup.exe”, escribir “server 10.0.2.2” y pulsar Enter para asegurar que se consulta exactamente a nuestro servidor DNS:



14. En “nslookup.exe”, escribir “srv2022.elorrieta.local” y pulsar Enter para probar el funcionamiento de la resolución directa:

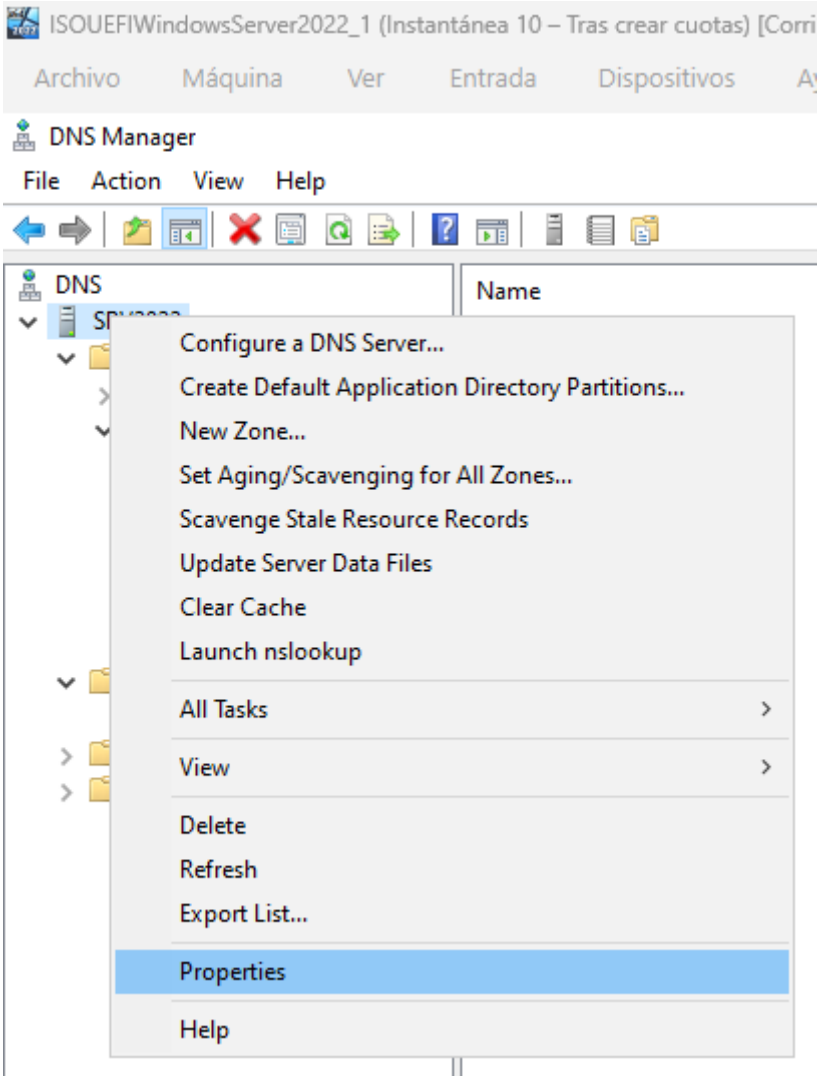
```
> srv2022.elorrieta.local
Server:  srv2022.elorrieta.local
Address:  10.0.2.2

Name:    srv2022.elorrieta.local
Address:  10.0.2.2
```

15. En “nslookup.exe”, escribir “10.0.2.2” y pulsar Enter para probar el funcionamiento de la resolución inversa:

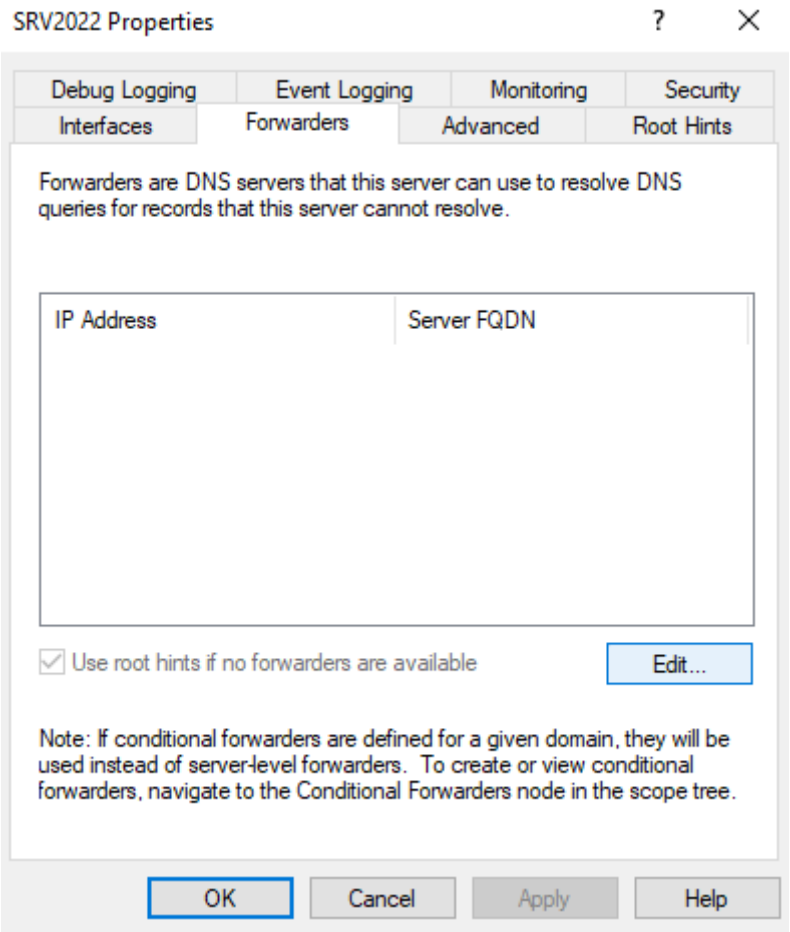
```
> 10.0.2.2
Server:  srv2022.elorrieta.local
Address:  10.0.2.2

Name:    srv2022.elorrieta.local
Address:  10.0.2.2
```

Configurar reenviadores

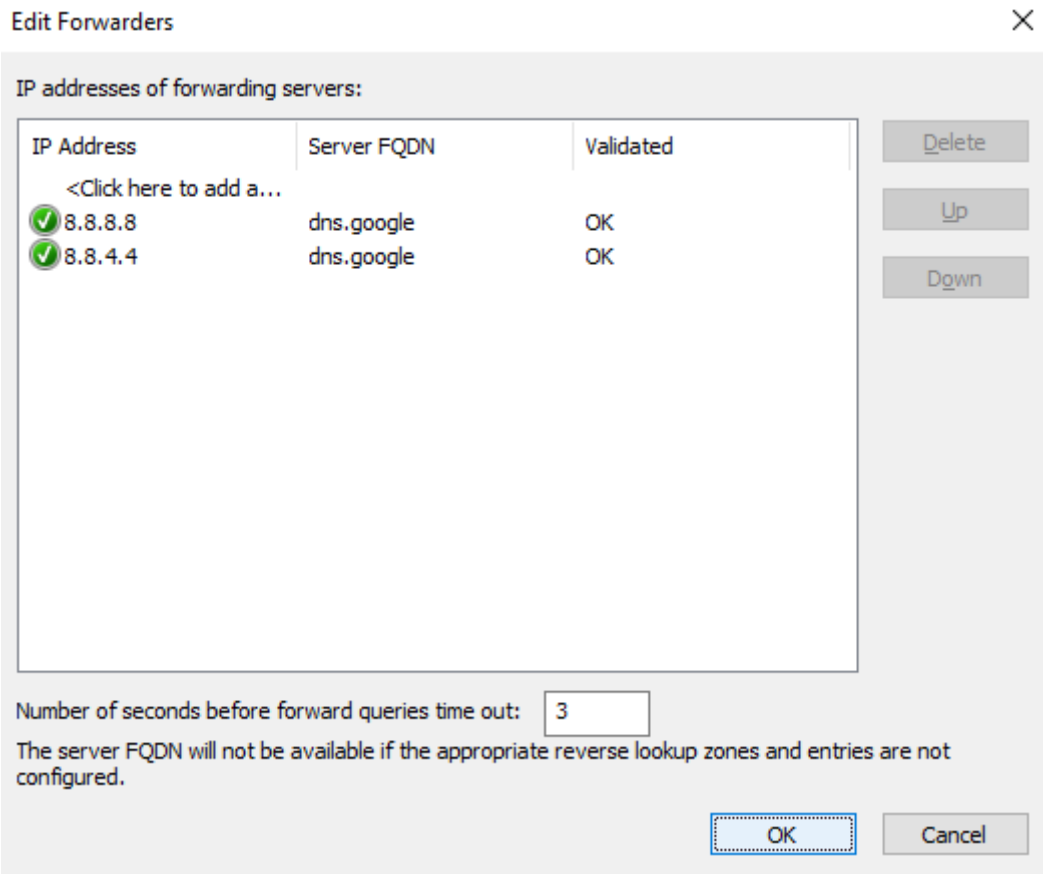
1. En “DNS manager”, en el panel lateral izquierdo, pulsar en “SRV2022”, click derecho en “SRV2022” y en el panel contextual pulsar en “Properties”:



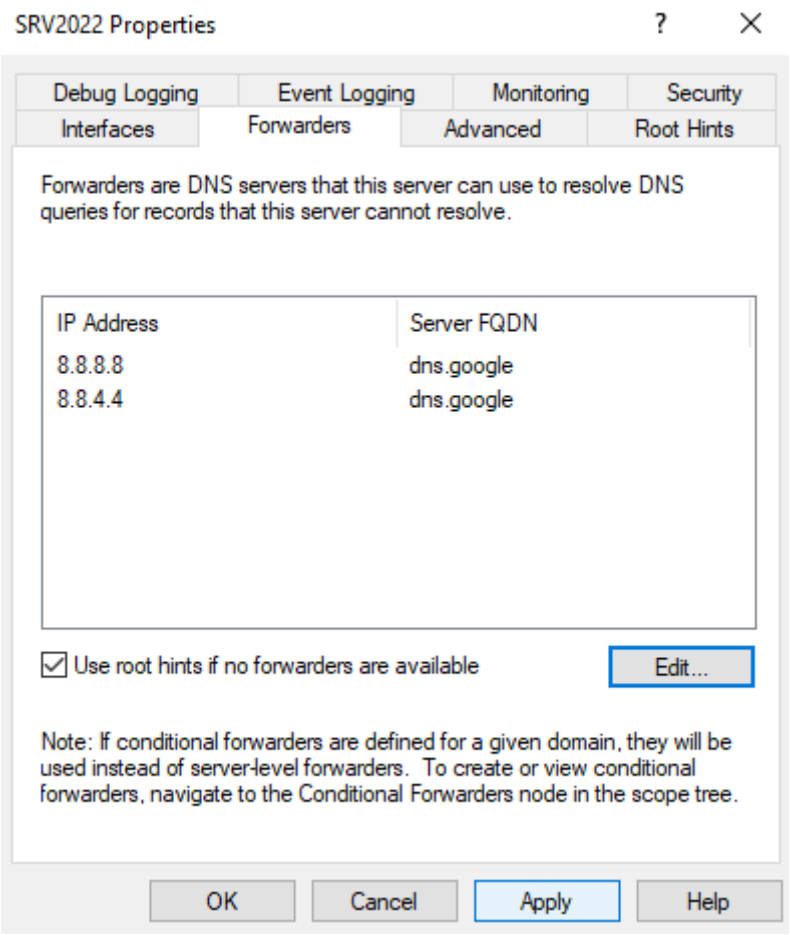
2. En “SRV2022 Properties” pulsar en “Forwarders”, y pulsar “Edit ...”:



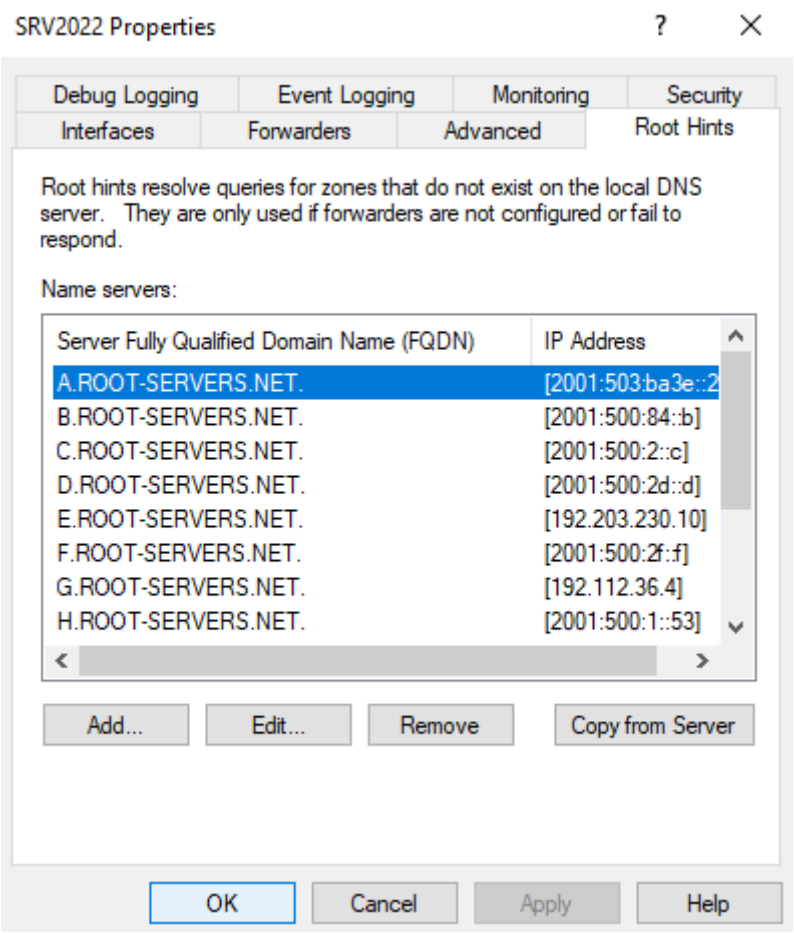
3. En “Edit Forwarders”, escribir las IPs “8.8.8.8” y “8.8.4.4”, dejarlas en ese orden mediante “Up y Down” y pulsar “OK”:



4. En “SRV2022 Properties” pulsar “Apply”:



5. En “SRV2022 Properties”, pulsar “Root Hints” y pulsar “OK”:


SRV2022 Properties
Debug Logging
Event Logging
Monitoring
Security
Interfaces
Forwarders
Advanced
Root Hints
Root hints resolve queries for zones that do not exist on the local DNS server. They are only used if forwarders are not configured or fail to respond.
Name servers:
Server Fully Qualified Domain Name (FQDN)
IP Address
A.ROOT-SERVERS.NET. [2001:503:ba3e::2
B.ROOT-SERVERS.NET. [2001:500:84::b]
C.ROOT-SERVERS.NET. [2001:500:2::c]
D.ROOT-SERVERS.NET. [2001:500:2d::d]
E.ROOT-SERVERS.NET. [192.203.230.10]
F.ROOT-SERVERS.NET. [2001:500:2f::f]
G.ROOT-SERVERS.NET. [192.112.36.4]
H.ROOT-SERVERS.NET. [2001:500:1::53]
Add...
Edit...
Remove
Copy from Server
OK
Cancel
Apply
Help

Realizar backups de zonas DNS

1. Abrir una terminal de cmd.
2. Escribir “dnscmd SRV2022 /ZoneExport elorrieta.local backup\elorrieta.local.dns” y pulsar Enter:

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>dnscmd SRV2022 /ZoneExport elorrieta.local backup\elorrieta.local.dns

DNS Server SRV2022 exported zone
  elorrieta.local to file %windir%\system32\dns\backup\elorrieta.local.dns on the DNS server
Command completed successfully.


C:\Users\Administrator>
```

3. Abrir una termina de PowerShell.
4. Escribir “Export-DNSServerZone elorrieta.local backup\elorrieta.local2.dns” y pulsar Enter:

Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Export-DNSServerZone elorrieta.local backup\elorrieta.local2.dns
PS C:\Users\Administrator>
```

5. Abrir una ventana del explorador de archivos:

ISOUEFIWindowsServer2022_1 (Instantánea 10 – Tras crear cuotas) [Corriendo] - Oracle VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

DNS Manager

File Action View Help

- DNS
  - SRV2022
    - Forward Lookup Zones
      - _msdcs.elorrieta.local
      - elorrieta.local
        - _msdcs
        - _sites
        - _tcp
        - _udp
        - DomainDnsZones
        - ForestDnsZones
    - Reverse Lookup Zones
      - 2.0.10.in-addr.arpa
    - Trust Points
    - Conditional Forwarders

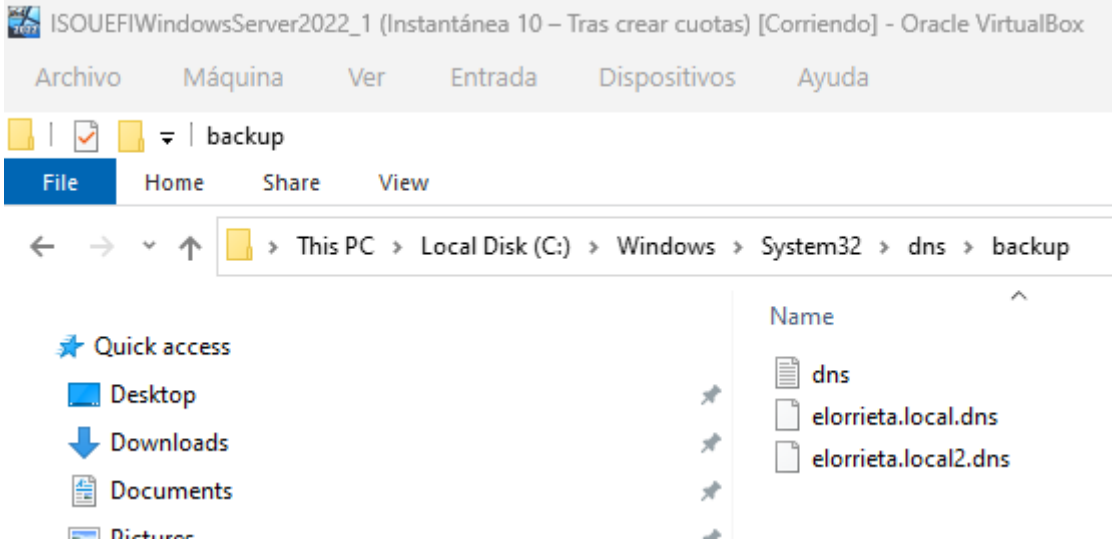
Name

- Forward Lookup Zones
- Reverse Lookup Zones
- Trust Points
- Conditional Forwarders
- Root Hints
- Forwarders

File Explorer

Type here to search

6. En la ventana del explorador de archivos, navegar hasta la ruta “C:\Windows\System32\dns\backup” y verificar que aparecen los ficheros “elorrieta.local.dns” y “elorrieta.local2.dns”:



7. Apagar la máquina virtual servidor.
8. Verificar que la máquina virtual servidor está apagada.
9. En el “Oracle VirtualBox Administrador”, tomar una instantánea/snapshot de la máquina virtual del servidor con nombre de la instantánea “Instantánea 11 – Tras configurar DNS”.